FEATURE 03 OF 14 | PRIORITY P0

Risk Scoring (0-100)

Priority	P0
Judging criteria served	Completeness (35) | Innovation (20)
Estimated effort	Small (~1-2 hrs) (rough estimate)
Depends on	Rule-Based Detection

Purpose

Turns the list of fired rules into one understandable number from 0 to 100. The score must be fully explainable: anyone can add up the points shown in the explanation panel and get the same total.

What it must do

- Score = sum of points of the rules that fired, **each rule counted once**, then capped at 100.
- Rules in the same group (OTP and password/PIN) count only the highest value of the group, so duplicate signals do not inflate the score.
- Weights, group caps and the 100 cap live in one config object so they can be tuned without code changes.
- The function returns the total plus a per-rule breakdown (rule id, points awarded) used by feature 6.
- Always display a note that the score is an estimate of detected warning signs, not a probability that fraud has occurred.

How it should work

- **Formula:** raw = sum(points of each fired rule, with group maximum applied); score = min(100, raw).
- **Your brief's example:** urgency 20 + account threat 15 + suspicious link 25 + OTP request 40 = 100 / 100.
- **Why group caps:** a message that says "send your OTP and PIN" asks for credentials once - not twice - so it should not score 80 on that alone. Both are still highlighted and explained; only the points are capped.
- **Optional safety override (config flag, off by default):** if any credential request fires, force the level to at least Medium and show a "Never share your OTP" banner even if the score is below 30. Off by default so the brief's thresholds stay exact; turn on if you want extra caution.
- Keep arithmetic in integers; no floating point or hidden multipliers.

UI / UX notes

- Large score number with a progress ring or bar, coloured by level (feature 4).
- Small "How is this calculated?" link opening the breakdown table.
- Show 'Score: 100 / 100' not just '100' so the scale is clear.

Edge cases and risks

- Nothing fires: score 0, level Low.
- Everything fires: capped at 100 - the sum is never shown above 100.
- OTP and password both fire: group cap applies, breakdown shows the second one as '0 (covered by group)' so the maths stays visible.

- Config accidentally sets a negative or non-number weight: validate config on load and fall back to defaults.

Acceptance tests

Try this	Expected result
Urgency + threat + link + OTP	100 / 100
OTP only	40 / 100 (Medium)
Urgency only	20 / 100 (Low)
OTP + password request	40 / 100 (group cap), both listed
Urgency repeated 5 times	20 / 100 - counted once

Demo tip for the judges

> Edit one weight in the config, refresh, and show the score change - proof the logic is configurable and transparent.